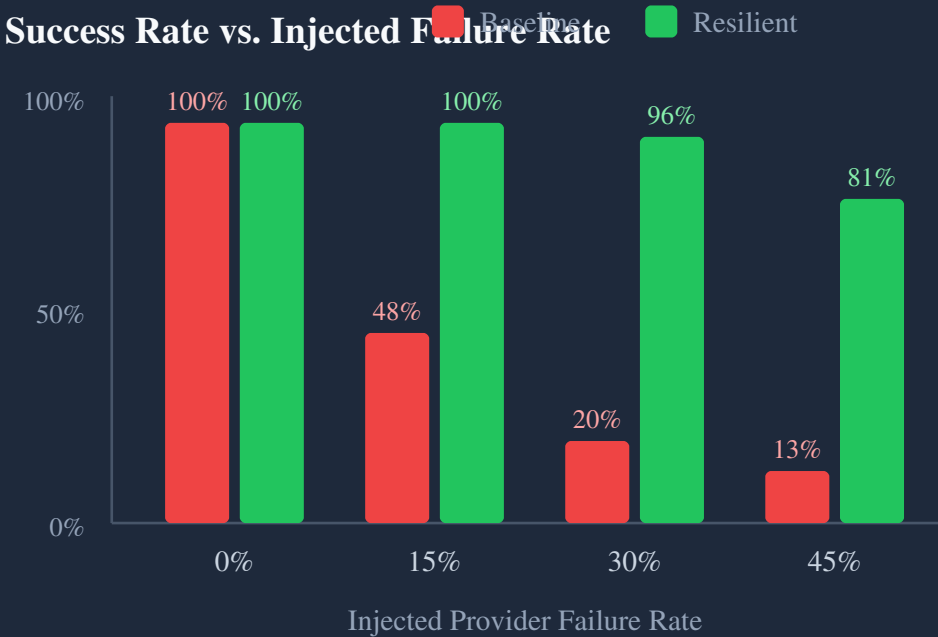


P15: Agent Resilience Under Fault Injection

Bounded Retries + Seeded Exponential Backoff + Circuit Breaker



Circuit Breaker Proof

Under Sustained 100% Outage

Load Shedding Ratio

80.0% Shed

Trip Threshold: 3 failures
Endpoint Calls: 3 of 15 sent
Fast-Failed Rejections: 12 dropped
Post-Cooldown State: CLOSED (probe ok)

Reliability vs. Cost Summary (Wilson Confidence Intervals)

- 0% Failures: 100% baseline [86%-100%] vs 100% resilient [86%-100%] (Cost: 0 extra calls, 0ms latency overhead).
- 30% Failures: Baseline degrades to 36% [20%-56%]; Resilient recovers to 96% [80%-99%] (Gain: +60%).
- Circuit Breaker trips to OPEN after 3 consecutive failures, preventing cascading downstream retry storms.